

Institute of Actuaries of India

Subject SP8 – General Insurance: Pricing

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

i)

- a. $RC/E \times A/RC + PC/E \times (P + W + S)/PC$ [0.5]
 Remainder = [0.5]
- b. Reported and Paid Frequency are RC/E and RC/E
- c. Defense cost severity (ALAE) = A/RC
- d. Punitive damage severity = P/PC
- e. Lost wages severity = W/PC
- f. Pain and Suffering severity = S/PC [2]

ii)

RC/E = [0.5 marks for each]

- a. Increasing compensation culture,
- b. Propensity of no-win-no-fee arrangements,
- c. Growth of claims management companies (placing of claims),
- d. Changes in regulations,
- e. Changes in economic conditions,
- f. Changes in policy terms, conditions, excesses, limits etc.

A/RC =

- a. inflation of legal costs

PC/E =

- a. court decisions
- b. emergence of latent claims

P/PC =

- a. court decision/inflation eg assessment of punitive damage

W/PC =

- a. salary inflation
- b. legislative changes such as structured settlements
- c. interest rate changes eg discount rate used for Ogden tables

S/PC =

- a. Court inflation in assessment of compensation for pain and suffering [6]

iii)

S/PC will show a one-time, sudden and measurable shift post this reform.

[0.5 marks for each valid consideration, 3 marks total]

- a. Past S/PC trend from experience period will not be indicative of the future.
- b. RC/E may show an uptick as people become aware of this news and try to file claims prior to effective date of reform.

- c. RC/E trends may therefore show an uptick but will not be reflective post effective date of reform
- d. A/RC may show a drop if most of the additional RC are frivolous claims
- e. Again, the trend for A/RC may not be indicative
- f. Any other valid considerations permissible.

[3]

[11 Marks]

Solution 2:

For ILFs to be consistent, they must be non-decreasing and increasing at a decreasing rate. [0.5]

- 1) In order to be non-decreasing – $2.7 \leq X \leq 5.5$ [0.5]
- 2) In order to be increasing at decreasing rate, we need to first compute the rates of increase which is difference in ILF/difference in Limit*1,00,000 [1.5]

Limit	ILF	Rate of Increase (per 1,00,000)
100,000	1	1.00
200,000	1.6	0.60
500,000	2.7	0.37
1,000,000	X	-
2,500,000	5.5	-
5,000,000	7	0.06
10,000,000	9	0.04

- 3) So, $0.37 > (X - 2.7)/5 > (5.5 - X)/15$ [1]
- a. So combining both gives $3.4 < X < 4.55$ [0.5]

[4 Marks]

Solution 3:

- i) $AIC1 = -2L1 + 2N$, $AIC2 = -2L2 + 4N$, $AIC1 = AIC2$ implies $L2 - L1 = N$. So log-likelihood should increase by at least the number of parameters N. [2]
- ii) We would reflect this variable as an offset term with values of $\log(1)$ and $\log(1.2)$ assuming the link function used is log. After this, we will not include this as an explanatory variable anymore in our GLM. The offset term ensures that the GLM routine doesn't fit parameters to the offset terms and fixes them to have final relativities of 1 and 1.2 while other variables are free to "pick up" or explain away any statistical deviation from the 20% constraint [2]
- iii) [0.5 marks for each point]
 - a. The fact that the AIC drops only marginally is of little consequence since it is not significant
 - b. I would put the error bars (± 2 sd around the estimates) to see how wide they are.
 - c. Given that AIC didn't drop much we would expect the error bars to be wide.
 - d. I would visually inspect if the three high parameter estimates and the two low one separately can be grouped into two groups.

- e. I would group them and review the AIC comparing it to the original model. If it reduces more, then we have done a good grouping. If it worsens then there is little chance of any better way to include this variable.
- f. If the AIC reduces by a similar magnitude, I would still consider including this factor however, I would investigate the significance of the standard errors of the estimates of the two groups. If the standard error of the estimates are relatively small then the factor might be worth keeping.
- g. In order to test the new grouping I would include a test with time to see if the pattern/difference holds over time and validate using a test set.
- h. I would also include a random factor to see if the difference in estimates that we find are purely by chance or not.
- i. If any of the 5 levels is unknown then I may exclude it from my grouping.
- j. I will consider the business context of how critical it is to either include or exclude the variable.
- k. I will enquire if this factor was included in prior studies and if so was a similar pattern observed then as well.
- l. If the importance has been reducing over time (since previous analysis) then this is explicable and it may be time to drop the variable altogether from the analysis. If this is a sudden and unexplained change, then we may have to investigate data integrity and other pre-processing steps.
- m. We should also investigate the exposure/claim distribution by the levels of this variable. If the majority of the exposures end up in one or the other group then there is little value in adding this grouped variable from a statistical point.

[6]

[10 Marks]

Solution 4:

- i) Comparison with validation data (out-of-sample or out-of-time) [0.5]
 - a. Lines showing actual vs predicted claims grouped by predicted claim bands on x-axis. [0.5]
 - b. Bars showing exposures in each band. [0.5]
 - c. If model is perfect, both lines would perfectly overlay. [0.5]
 - d. Systematic differences can indicate poor model fitting, for example if the line was consistently above or below the theoretical line. [0.5]
 - e. Departures of actual from expected are more concerning when such departures apply to a significant proportion of the portfolio i.e where exposure bars are high. [0.5]
- ii) Lift curves on validation data [0.5]
 - a. Useful for comparing 2 models of different forms. [0.5]
 - b. Rank all policies in the validation dataset in order of expected experience and then group the policies into bands of equal exposure based on this ranking. [0.5]
 - c. Actual experience for each group can then be calculated and displayed as a curve. [0.5]

- d. The steeper the curve, the more effective the model is at distinguishing between high and low experience. [0.5]
- e. Difference between frequency of lowest group and the highest group is the lift. [0.5]
- iii) Gains curve [0.5]
 - a. Data is sorted high to low according to the fitted model values [0.5]
 - b. Chart shows cumulative values from the fitted model and the observed values from the data. [0.5]
 - c. Statistical measure of the lift produced by the model is called the Gini coefficient [0.5]
 - d. This is the area enclosed by the model curve and the diagonal line as a ratio of the triangle above the diagonal. [0.5]
 - e. Diagonal line represents reference model or a random model [0.5]
 - f. Gini is a measure of statistical dispersion than can range from 0 to 1. [0.5]
 - g. Higher the Gini coefficient, the more predictive the model. [0.5]

[10 Marks]

Solution 5:

- i) Differences [0.5 marks for each]
 - a. Although the formula are very different, they will produce approximately the same credibility weights if the full credibility standard for classical is 7 to 8 times larger than the Bayesian credibility factor k
 - b. The most significant difference between the two models is that Bayesian cred. Never reaches 1.
 - c. Either model can be effective at improving the stability and accuracy of estimates
 - d. For a particular application, actuary can choose the model appropriate to the data and the goals.
 - e. When estimates of $E(s^2)$ and $\text{var}(m)$ are available, Bayesian credibility generates the most accurate insurance rates and forms the basis of most experience rating plans.
 - f. Classical credibility can be used if these estimates are unknown or difficult to calculate and is often used in the calculation of overall rate increases.
 - g. Larger insureds with good claims experience usually want their premium to be based solely on their own experience and when full credibility standards are met, classical credibility can do so.
 - h. Classical credibility is simpler to work with and easier to explain to non-actuaries. [4]
- ii) Desirable qualities –
 - a. Accuracy as a predictor of next year's mean loss costs (that is, low variance around next year's mean loss costs). Our selected 3 year mean should be adjusted for any expected trend. In that case is expected to be an accurate predictor. [0.5]
 - b. Unbiasedness as a predictor of next year's mean expected losses (that is, the differences between the predictor and the subsequent loss costs should average out near zero). This is expected. [0.5]

- c. Independence from the base statistic. Unless credit cards which were compromised in the past 3 years are more likely to be compromised again in this year, the measure is independent. [0.5]
 - d. Availability of data. Confirmed, since prior data should be available from bank partner for pricing anyways. [0.5]
 - e. Ease of computation. Confirmed. [0.5]
 - f. Explainable relationship to the loss costs of the class or individual being rated. Confirmed. [0.5]
 - g. Ease of explanation to managers and customers. Confirmed. [0.5]
 - h. Time vs accuracy. Confirmed. [0.5]
 - i. Chance of error in calculating the complement. Confirmed. [0.5]
 - j. The sources of error in the complement. [0.5]
- [5]**
[9 Marks]

Solution 6:

Major consideration is assessing likely overall loss ratio and what level of ceding commission can be offered whilst still retaining an acceptable profit for the reinsurer.

For Loss ratio

[0.5 marks for each]

- a. For assessing overall loss ratio, similar exercise to reserve review
- b. Triangulated premium, paid and incurred data needed. Since, proportional covers are usually on a risks incepting basis, so triangles should be on underwriting year basis. Separate triangles for each line.
- c. Rate change information over the time covered by the data split by LOB.
- d. Estimates of premium income to be written and rate changes for the period of cover being priced.
- e. Info on changes in mix of business, policy terms and conditions, over the period.
- f. Info on cat exposure.
- g. Next, appropriate pricing method to project to ultimate, calculate resulting loss ratio, trend and on-level premium.

For ceding/profit commission terms

[0.5 marks for each]

- a. If no profit commission, only check if 100-loss ratio-ceding commission leaves enough to cover the reinsurers' expenses and profit.
- b. Reinsurer may use the loss ratio distribution to identify the probability of the reinsurer making a loss. If this is too high, then the reinsurer should reduce the ceding commission even though the mean outcome is satisfactory.
- c. Under sliding scale commission, then loss ratio distribution should be used to calculate the distribution of the reinsurer's financial position.

- d. In such cases, reinsurer will often have requirements related to a prob. Of making a loss of no more than x%, making a loss of y% or more but no more than z%, or a combination.
- e. Reinsurer can adjust the ceding comm. Scales or profit comm. Until these requirements are met.
- f. It is common for the cedant or their broker to indicate the terms they would prefer, and if these meet the reinsurer's requirements no adjustment will be needed. If not, reinsurer has to negotiate.

[6 Marks]

Solution 7:

- i) Various types of reserves maintained by a GI company

[0.5 marks each]

- Reserves for Outstanding Reported claims
- Unearned Premium Reserves
- Unexpired Risk Reserve
- Claim Equalization reserve
- Catastrophe Reserve

[2]

- ii) Components of Outstanding claim reserves

[0.5 marks each]

- Reserves for Outstanding Reported claims: This is the estimated reserve needed to settle the claims that the company knows about at the accounting date.
- Reserves for IBNR Claims: The IBNR reserve is needed to cover the claim payments for incidents which have happened, but have not been reported to the insurance company
- Reserves for IBNER Claims: the IBNER reserve is needed to cover the expected increase (or decrease) in estimates for reported claims
- Reserves for reopened Claims: this is an additional reserve which may be explicitly shown to allow for claims that the insurance company treats as being fully settled, but which might one day require further payments.
- Reserves for claim handling expenses: While settling claims in each of the above classes, the company will incur some additional expenses. The reserves for these expenses may be held separately.

[2]

[4 Marks]

Solution 8:

The recovery shall be as follows:

Scenario 1					Recovery			
Vehicle Type	Gross Premium	Net Premium	Gross Loss	Loss Ratio	Quota @20%	Loss Corridor	XoL Recovery	Recovery
Commercial	500	400	500	100%	100	15		85
Private	600	480	480	80%	96			96
Total	1100	880	980	89%	196	15	0	181

[0.5 marks for QS recovery each, 0.5 marks for Loss Corridor and 0.5 mark for Total Recovery]

Scenario 2					Recovery			
Vehicle Type	Gross Premium	Net Premium	Gross Loss	Loss Ratio	Quota @20%	Loss Corridor	XoL Recovery	Recovery
Commercial	500	400	600	120%	120	15		
Private	600	480	750	125%	150			
Total	1100	880	1350	123%	270	15	24	279

[0.5 marks for QS recovery each, 0.5 marks for Loss Corridor and 0.5 mark for Total Recovery]

[4 Marks]

Solution 9:

[2 marks for each regulation and justification]

- a. **Regulation:** Simple and Standard Products e.g. standard terms and conditions
Justification: This allows customers to compare insurers' products more easily, without surprises. It also reduces the possibility for insurer to exploit customers' lack of financial knowledge by introducing larger excesses or exclusions to offer seemingly similar products at lower cost.
Check/test: Require products offered to low-income earners to go through the regulator for approval.
- b. **Regulation:** Impose limits on premium level.
Justification: Low-income market may be unable to afford high premiums, and are also not as able to discern whether premiums are fair or not. It is ultimately in the interest of building trust and confidence in the insurance industry.
Check/test: Require approval of premium rates by the Regulator. Check Loss Ratios
- c. **Regulation:** Maximum claim settlement duration
Justification: Increase value of insurance and increase trust in insurers. The low-income market will need prompt settlements as cash flow may be a problem for them.
Check/test: Require company to file a report of claims settlement times, including aggregate figures such as average settlement time and details of individual claims that went over the settlement limit, with comments on why this was the case.

[6 Marks]

Solution 10:

- i) **Insurable Interest:** For a risk to be insurable:

- a. The policyholder must have an interest in the risk being insured, to distinguish between insurance and gambling
- b. A risk must be of a financial and reasonably quantifiable nature
- c. The amount payable by the insurance policy in the event of a claim must bear some relationship to the financial loss incurred

[1]

- ii) **Linking the insurability of the vehicles with above stated points of insurable interest**

- a. It is reasonable to assume that owners of unique vehicles have an interest in the risk as it is unlikely that they will be able to replace their vehicles should something happen to them. [0.5]
- b. However, during a recession, it may be less likely that individuals will be able to sell such cars, making insurance fraud a possibility as car owners aim to get cash. [1]

- c. This may be difficult to mitigate as the insurance payout will be unlikely to achieve indemnity due to the limited nature of the cars, hence will pay out in cash. [0.5]
- d. The vehicles should have a market price, or a reference point with which to estimate the sum insured. [0.5]
- e. Being property insurance, there will naturally be limits on the payout, which should coincide with the sum insured. [0.5]
- f. In general, motor insurance claims are of low enough probability to insure (evidenced by the extent of motor insurance). [0.5]
- g. Claims on unique vehicles will likely be lower as such vehicles are often for investment purposes and are unlikely to be driven as much. [0.5]
- h. Depending on the definition of “unique vehicles”, certain categories of vehicle may have higher claim frequencies – special cases will need to be considered separately. [0.5]
- i. The Company may be able to reinsure and gain expertise from the reinsurer. [0.5]

[5]

[6 Marks]

Solution 11:

i) Differences resulting from the types of business concerned include:

- The insurer selling motor and Home insurance should have much larger volumes and better data quality than the insurer selling professional indemnity (PI) business. [1]
- Both for Motor and Home insurance, the Company should be able to identify potential data errors through built in error checking validation when data are captured. [0.5]
- Direct business is also transacted through the internet, which enables the company to tailor make the input they would require and have the user enter this information at the point of underwriting. [0.5]
- The Professional Indemnity business is sold mainly through specialized agents and hence might not have complete control on the underlying data [0.5]
- ...i.e. the company might have multiple brokers writing this business which could lead to less consistency with the data. [0.5]
- Also, the agents might only be sharing Bordeaux data at an aggregate level. [1]
- The system design for Motor and Home should be able to capture different rating factors used for pricing across policy and claims data. However, in case of PI the system is expected to be relatively simple with small number of rating factors. [1]

[5]

ii) The aggregate claims arising from each portfolio have a compound Poisson distribution. Let S_1 be the aggregate claims arising from Motor insurance portfolio and let S_2 be the aggregate claims arising from Home insurance portfolio.

$$S_1 = X_1 + X_2 + \dots + X_N \text{ where } N \text{ is Poisson (20) and } X \text{ is Gamma (20, 5)}$$

$$S_2 = Y_1 + Y_2 + \dots + Y_M \text{ where } M \text{ is Poisson (5) and } Y \text{ is Gamma (10, 2)}$$

The aggregate claim amount from all policies is $S_1 + S_2$, which will have a compound Poisson distribution with Poisson Parameter 25. [0.5]

From the Tables, the mean, second and third non-central moment for the gamma distribution are given by:

$$E(X) = \frac{\alpha}{\lambda}$$

$$E(X^2) = \frac{\alpha(\alpha+1)}{\lambda^2}$$

$$E(X^3) = \frac{\alpha(\alpha+1)(\alpha+2)}{\lambda^3}$$

For S_1 , we have:

$$E(S_1) = \lambda E(X) = 20 * 20 / 5 = 80$$

$$\text{Var}(S_1) = \lambda E(X^2) = 20 * (20 * 21) / 5^2 = 336$$

$$\text{Skew}(S_1) = \lambda E(X^3) = 20 * (20 * 21 * 22) / 5^3 = 1478.4$$

[0.5]

Where $\text{Skew}(S_1)$ denotes the third central moment of S_1 . Similarly for S_2 , we have:

$$E(S_2) = \lambda E(X) = 5 * 10 / 2 = 25$$

$$\text{Var}(S_2) = \lambda E(X^2) = 5 * (10 * 11) / 2^2 = 137.5$$

$$\text{Skew}(S_2) = \lambda E(X^3) = 5 * (10 * 11 * 12) / 2^3 = 825$$

[0.5]

If we assume that all risks are independent, then the means, variances and skewnesses of S_1 and S_2 are additive and we can find first three moments of S :

$$E(S) = E(S_1) + E(S_2) = 80 + 25 = 105$$

$$\text{Var}(S) = \text{var}(S_1) + \text{var}(S_2) = 336 + 137.5 = 473.5$$

$$\text{Skew}(S) = \text{skew}(S_1) + \text{skew}(S_2) = 1478.4 + 825 = 2303.4$$

[0.5]

[5]

[10 Marks]

Solution 12:

[0.5 mark each]

Allocating expenses

- Accurate data is required for an accurate expense analysis.
- The data needs to be split by activity/function, direct/indirect expense etc.
- We need to ascertain whether costs depend on volume of business, premium size or some other element.
- However, completely accurate data may not be available at this level.
- Many functions are central or there is no obvious way of splitting costs which creates additional difficulty.
- Approximations may be needed, such as time-sheet sampling and pro-rating by headcount/floor space, infrastructure used.
- These allocations can be inaccurate and become out-of-date quite quickly.

Loading into the rates

- If the expense splits are not accurate, the modelled costs may not reflect actual costs if the business mix from various distribution channel changes.

- The expenses are bound to be different for different channels. Some channels may be having high fixed/initial cost e.g. Online/Direct channel and few channels may be having high variable cost e.g. Agency channel.
- This may result into different per policy expense cost as business volume changes.
- However, this may mean that initial business sourced through online/direct channel may be uncompetitive or loss making.
- So, business from other channels may need to subsidise online/direct channel business.
- This further requires assumptions on sales and persistency, which introduce uncertainty.

[4 Marks]

Solution 13:

i) Investigations around following parameters need to be done to ascertain the reason for deterioration:

- a. Premium Rates adequacy
- b. Portfolio Movements
- c. Business Mix – product mix and distribution channel mix
- d. Persistency
- e. Competition
- f. Policy Terms and conditions
- g. Claims frequency for both Motor OD and Motor TP
- h. Claims severity for both Motor OD and Motor TP
- i. Court Awards for Injury and Deaths
- j. Inflation including Wage inflation
- k. Proportion of claims split by Injury and Death
- l. Effect of Large claims / Aggregation of claims
- m. Customer behavior
- n. Other sensible comments

Using the investigation results:

- a. Correcting the premium rates if found unviable
- b. Correcting the unfavorable portfolio movement like low business volumes, high cancellation rates etc.
- c. Correcting the business mix – diverting the business towards more profitable products and through low cost channels
- d. Taking immediate action to improve persistency
- e. Comparing the product features with that of competitors and also the pricing
- f. Customer behavior towards claiming may be different for 1 year and 3-year policy. Making changes in the policy terms to control the same.

[10]

ii) a. **Risks associated with long term policies**

- a. Premium collected at the start may not be adequate

- b. There might be type of claims which were not anticipated at the time of pricing i.e. latent claims
- c. Change in customer behavior may be different from anticipated
- d. High mid-term cancellation may lead to not gaining the expected renewal retention rates
- e. Anti-selection
- f. Claim incidence rate may be higher than priced for
- g. Claim inflation may be higher than priced for
- h. Changes in regulations

b. Mitigating the risks

- a. Experience may be gained initially by selling the policies as pilot run
- b. Continuously monitoring the business and taking corrective actions as the need arises
- c. Taking expertise from reinsurers

[6]

[16 Marks]
